GE Healthcare

Technical Publications

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Rev.9

CE 0459

E8C Type Probe/E8C-RS Type Probe User Manual

Operating Documentation

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Regulatory Requirement

The E8C Type probe complies with regulatory requirements of the following European Directive 93/42/EEC concerning medical devices for the ultrasound system.

CE 0459

The E8C-RS Type Probe complies with regulatory requirements of the following European Directive 93/42/EEC concerning medical devices for the ultrasound system.

CE 0459

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EC REP

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Revision History

Reason for Change

REV	DATE	REASON FOR CHANGE
Rev. 0	June 1st, 2001	Initial Release
Rev. 1	Aug. 10, 2001	Add the applicable system
Rev. 2	May 14, 2002	Add E8C-RS Probe
Rev. 3	Dec. 25, 2002	Add the applicable system
Rev. 4	May 27, 2003	Add the manufacturer and the applicable system
Rev. 5	Feb. 26, 2004	Add the description of a removal tool and Cidex Opa
Rev. 6	April 17, 2008	Remove the applicable system description from Page 1.
Rev. 7	Nov. 6, 2008	Revise Eu Rep and List 3 of Storing the probe.
Rev. 8	July. 17, 2009	Add EURep symbol and Alter the manufacturer name.
Rev. 9	Dec. 25, 2009	Remove system name from Regulatory requirement description.

List of Effective Pages

PAGE NUMBER	REVISION NUMBER	PAGE NUMBER	REVISION NUMBER
Title Page	Rev.9	Table of Contents	Rev.9
Revision History	Rev.9	1 thru 16	Rev.9

Please verify that you are using the latest revision of this document. Information pertaining to this document is maintained on MyWorkshop/ePDM (GE Healthcare Electronic Product Data Management). If you need to know the latest revision, contact your distributor, local GE Sales Representative or in the USA call the GE Ultrasound Clinical Answer Center at 1 800 682 5327 or 1 262 524 5698.

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Table of Contents

Instructions

General

The probe is 6.5MHz micro-convex device for conducting transvaginal scan.

Composition

1. Probe body, including:
   - Head-Assy
   - Grip
   - Carrying Case
2. Operation manual
3. Filling remover tool

Specifications

Scanning Method	Micro-convex, 11R
Frequency	6.5MHz
Field of View	133 degrees
Number of elements	128
Weight	Approximately 140g (not including the cable and connector)

Outside Dimensions and Construction

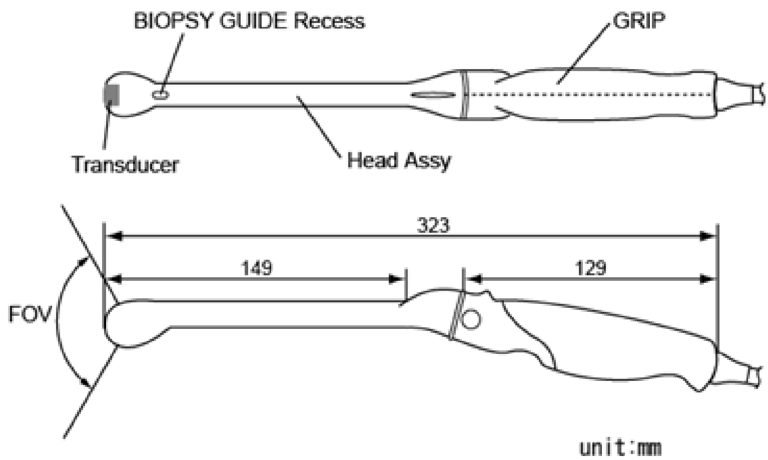

Figure -1.　Probe dimensions

Operation

Refer to the section entitled Connecting the probe in the ultrasound system user manual and connect the probe to the probe connector provided on the control console.

WARNING

Ultrasound transducers can easily be damaged by improper handling and by contact with certain chemicals. Failure to follow these precautions can result in serious injury and equipment damage.

- Do not immerse the probe into any liquid beyond the level specified for that probe. Never immerse the transducer connector or probe adapters into any liquid.
- Avoid mechanical shock or impact to the transducer and do not apply excessive bending or pulling force to the cable.
- Transducer damage can result from contact with inappropriate coupling or cleaning agents:
  - Do not soak or saturate transducers with solutions containing alcohol, bleach, ammonium chloride compounds or hydrogen peroxide
  - Avoid contact with solutions or coupling gels containing mineral oil or lanolin
  - Avoid temperatures above 60°C.
- Inspect the probe prior to use for damage or degeneration to the housing, strain relief, lens and seal. Do not use a damaged or defective probe.

How to install

Before installing the attachment, the filling must be removed from the probe head where the biopsy guide attachment will be installed. This is only for first time installation. See Figure 2.

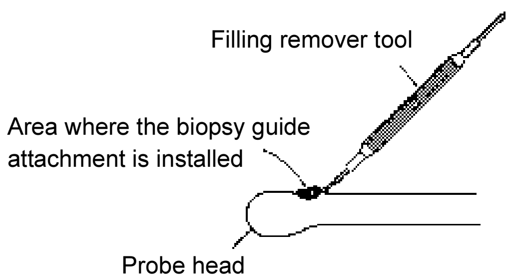

Figure -2.   Remove the filling

1. Fill the probe sheath with acoustic coupling gel as shown in Figure 3.

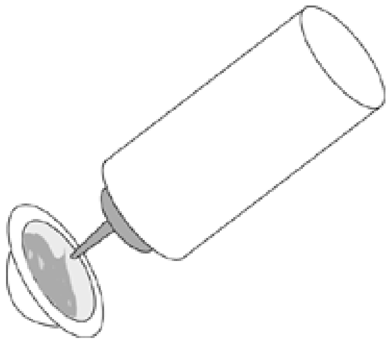

Figure -3. Ultrasound gel placement

2. Place the gel-filled probe sheath on the transducer face and bring it up the probe body so that the adjustment knob of the probe is completely covered with the probe sheath as shown in Figure 4.

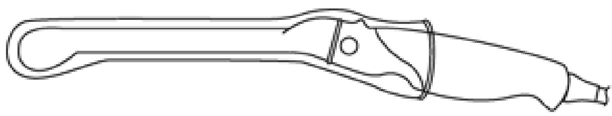

Figure -4. Probe sheath application

CAUTION

Do not use pre-lubricated condoms as a sheath. In some cases, they may damage the probe. Lubricants in these condoms may not be compatible with probe construction.

CAUTION

DO NOT use an expired probe sheath. Before using probe sheaths, verify whether the term of validity has expired.

3. Apply gel to outer surface of theprobe sheath at the transducer location for scanning.

4. Figure 5 shows the relationship between the displayed image direction during an axial scan compared to the probe and probe position.

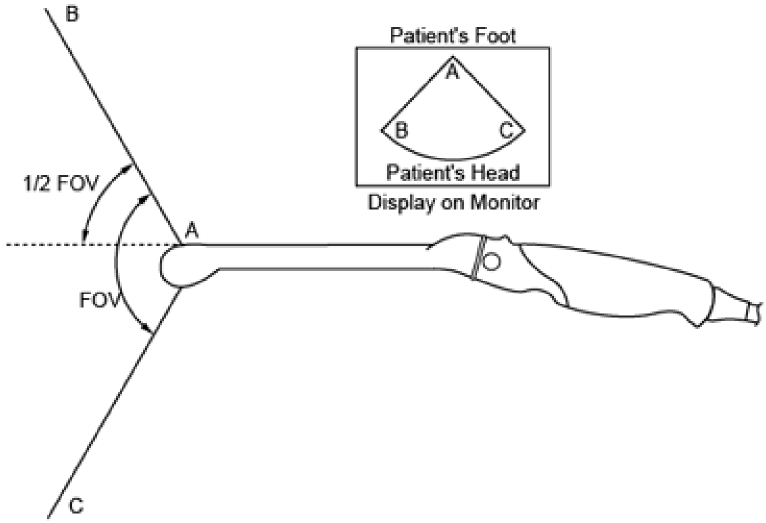

Figure -5. Scan plane orientation

Cleaning and Disinfecting the Probe

The probe is driven with electrical energy that can injure the patient or user if live internal parts are contaced by conductive solution.

CAUTION

DO NOT immerse the probe into any liquid beyond the level indicated by the immersion level in Figure 6 on page-9. Never immerse the probe connector or probe adaptors in any liquid.

Probe Cleaning Process

Avoid cross contamination. Follow all infection control policies established by your office, department or hospital as they apply to personnel and equipment.

1. Disconnect the probe from the ultrasound console and remove all coupling gel from the probe by wiping with a soft cloth and rinsing with flowing water.
2. Wash the probe with mild soap in lukewarm water. Scrub the probe as needed using a soft sponge, gauze, or cloth to remove all visible residue from the probe surface. Prolonged soaking or scrubbing with a soft bristle brush (such as a toothbrush) may be necessary if material has dried onto the probe surface.
3. Rinse the probe with enough clean potable water to remove all visible soap residue.
4. Air dry or dry with a soft cloth.

CAUTION

DO NOT clean any portion of the probe with methanol, ethanol, isopropanol, or any other alchol based cleaner. Such substances can cause irreparable damage to the probe.

Disinfecting probes

Daily Weekly Monthly Other

After Each Use

Ultrasound probes can be disinfected using liquid chemical germicides. The level of disinfection is directly related to the duration of contact with the germicide. Increased contact time procedures a higher level of disinfection. 2% Glutaraldehyde-based solutions have been shown to be very effective for this purpose. Cidex and Cidex Opa are the germicide that have been evaluated for compatibility with the material used to construct the probes.

Refer to the Probe Care Card or http://www.gehealthcare.com/usen/ultrasound/products/probe_care.html.

CAUTION

In order for liquid chemical germicides to be effective, all visible residue must be removed during the cleaning process.Thoroughly clean the probe as described earlier before attempting disinfection.

1. Prepare the germicide solution according to the manufacturer's instructions. Be sure to follow all precautions for storage, use and disposal.
2. Place the cleaned and dried probe in contact with the germicide for the time specified by the germicide manufacturer. High-level disinfection is recommended for surface probes and is required for endocavitary and intraoperative probes (follow the germicide manufacturer's recommended time).

   Probes for neuro surgical intra-operative use must not sterilized with liquid chemical sterilants because of the possibility of neuro toxic residues remaining on the probe. Neurological procedures must be done with the legally marketed, sterile, pyrogenfree probe sheaths.
3. After removing from the germicide, rinse the probe following the germicide manufacturer's rinsing instructions. Flush all visible germicide residue from the probe and allow to air dry.

CAUTION

CREUTZFIELD-JACOB DISEASE

Neurological use on patients with this disease must be avoided. If a probe becomes contaminated, there is no adequate disinfecting means.

Biological Hazard

High Level Probe Disinfection

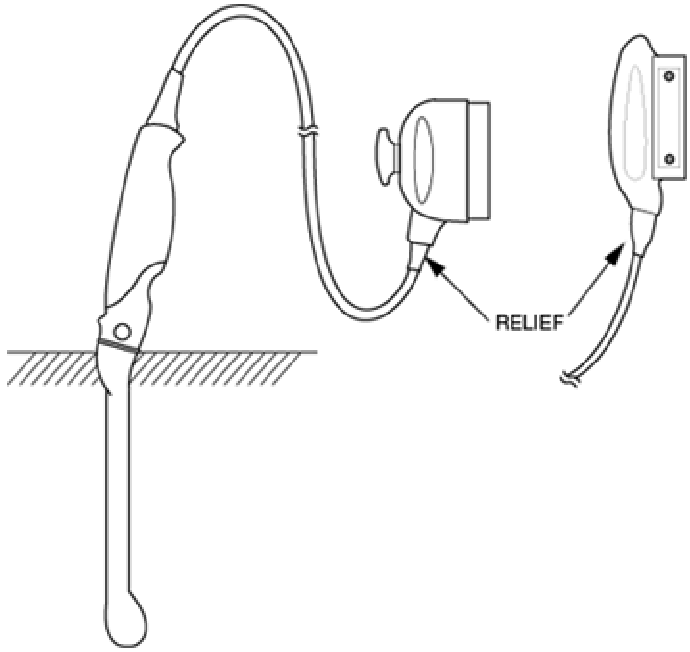

Figure -6. Probe immersion level

CAUTION

- DO NOT attempt to disinfect or sterilize the probe with an autoclave or steam sterilization.
- DO NOT clean the probe by immersing in an alcohol solution.
- DO NOT soak, disinfect, or sterilize the connector and relief.
- DO NOT immerse the grip part of the probe.

Coupling gels

WARNING

Do not use unrecommended gels (lubricants). They may damage the probe and void the warranty.

Applying

In order to assure optimal transmission of energy between the patient and probe, a conductive gel or couplant must be applied liberally to the patient where scanning will be performed.

CAUTION

Do not apply gel to the eyes. If there is gel contact to the eye, flush eye thoroughly with water.

Precautions

Coupling gels should not contain the following ingredients as they are known to cause probe damage:

- Methanol, ethanol, isopropanol, or any other alcohol-based product
- Mineral oil
- Iodine
- Lotions
- Lanolin
- Aloe Vera
- Olive Oil
- Methyl or Ethyl Parabens (para hydroxybenzoic acid)
- Dimethylsilicone
- Polyether glycol based

Storing the Probe

1. After cleaning and disinfection, store the probe in it's carrying case for protection.
2. Store the probe even when traveling for short distances in its carrying case or in the system probe holder.
3. The probe should be stored or transported within the parameters outlined below.

Table 1: 2D Probe Environmental Requirements

	Operational	Storage	Transport
Temperature	10° - 40° C 50° - 104° F	-10° - 60° C 14° - 140° F	-40° - 60° C -40° - 140° F
Humidity	30 - 85% non-condensing	30 - 90% non-condensing	30 - 90% non-condensing
Pressure	700 - 1060hPa	700 - 1060hPa	700 - 1060hPa

The following maintenance schedule is suggested for the system and probes to ensure optimum operation and safety..

Table 2: Planned Maintenance Program

Do the Following	Daily	After Each Use	As Necessary
Inspect the probes	X		
Clean the probes		X	
Disinfect the probes		X	
Sterilize probes			X

Contact Information

Contacting GE Healthcare Ultrasound

For additional information or assistance, please contact your local distributor or the appropriate support resource listed on the following pages:

INTERNET http://www.gehealthcare.com

http://www.gehealthcare.com/usen/ultrasound/products/probe_care.html

USA GE Healthcare TEL: (1) 800-437-1171
Ultrasound Service Engineering FAX: (1) 414-721-3865
9900 Innovation Drive
Wauwatosa, WI 53226

Clinical Questions For information in the United States, Canada, Mexico and parts of the Caribbean, call the Customer Answer Center
TEL: (1) 800-682-5327 or (1) 262-524-5698

In other locations, contact your local Applications, Sales or Service Representative.

Service Questions For service in the United States, call GE CARES

TEL: (1) 800-437-1171

In other locations, contact your local Service Representative.

Accessories Catalog Requests To request the latest GE Accessories catalog or equipment brochures in the United States, call the Response Center

TEL: (1) 800-643-6439

In other locations, contact your local Applications, Sales or Service Representative.

Contacting GE Healthcare Ultrasound (continued)

Placing an Order

To place an order, order supplies or ask an accesory-related question in the United States, call the GE Access Center

TEL: (1) 800-472-3666

In other locations, contact your local Applications, Sales or Service Representative.

CANADA

GE Healthcare TEL: (1) 800-664-0732
Ultrasound Service Engineering
9900 Innovation Drive
Wauwatosa, WI 53226
Customer Answer Center TEL: (1) 262-524-5698

LATIN & SOUTH AMERICA

GE Healthcare TEL: (1) 262-524-5300
Ultrasound Service Engineering
9900 Innovation Drive
Wauwatosa, WI 53226
Customer Answer Center TEL: (1) 262-524-5698

EUROPE

GE Ultraschall TEL: 0130 81 6370 toll free
Deutschland GmbH & Co. KG TEL: (33) 130.831.300
Beethovenstrasse 239 FAX: (49) 212.28.02.431
Postfach 11 05 60
D-42655 Solingen

ASIA

GE Ultrasound Asia (Singapore) TEL: 65-291 8528
Service Department - Ultrasound FAX: 65-272-3997
298 Tiong Bahru Road #15-01/06
Central Plaza
Singapore 169730

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GE Healthcare Japan Corporation TEL: (81) 42-648-2910
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BRAZIL	GE Sistemas Medicos TEL: 0800-122345 Av Nove de Julho 5229 FAX: (011) 3067-8298 01407-907 Sao Paulo SP
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LUXEMBOURG	TEL: 0800 2603 toll free

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POLAND	GE Medical Systems Polska TEL: +48 2 625 59 62 Krzywickiego 34 FAX: +48 2 615 59 66 P-02-078 WARSZAWA
PORTUGAL	GE Medical Systems Portuguesa S.A. TEL: 05 05 33 7313 toll free Rua Sa da Bandeira, 585 FAX: +351 2 2084494 Apartado 4094 TLX: 22804 P-4002 PORTO CODEX
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SWITZERLAND	GE Medical Systems (Schweiz) AG TEL: 155 5306 toll free Sternmattweg 1 FAX: +41 41 421859 CH-6010 KRIENS
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